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I'm working on a potential project that explores using AI to assist the validation of results in clinical trials papers. The motivation is that, when I serve as a reviewer, I often see some potential issues in the presented statistical results. An example is that some confidence intervals do not look correct, which do not look symmetric around the point estimates (for ratio measures such as odds ratio, then recall that the symmetry should be on the log scale), or point estimates may be outside the confidence intervals. I'm not saying that the results are intentionally fake; they could be due to typos (e.g., copying from some earlier contents and the authors forgot to update).

I'm thinking about a project of systematically assessing the performance of AI in such issue detection for published papers, e.g., clinical trial papers published in certain journals in certains.

Currently, I'm trying to do some preliminary research. How do you think about this idea?

I found a recent paper from JAMA; attached please find the main manuscript along with all its supplemental files. Could you please carefully assess all documents and let me know if you spot any issues? If you do spot any issues, please clearly give the location and the reason why it's an issue. Thanks!

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Thought for 5m 24s

I think the project idea is strong and timely, especially if framed as **AI-assisted statistical consistency checking**, not as detecting “fake” results. A good design would combine deterministic checks with AI extraction: AI identifies tables, outcomes, estimands, and relevant scales; then rule-based checks verify arithmetic and internal consistency. The key is to predefine an error taxonomy, such as numerator/denominator percentage mismatch, point estimate outside CI, CI symmetry on the correct scale, P value–CI inconsistency, direction inconsistency, protocol/SAP mismatch, and cross-table discrepancies.

For the uploaded TRAUMOX2 paper, I checked the main article and supplements with a focus on issues that can be verified from reported aggregate results. I did **not** find an obvious problem with the headline primary result: the main article and eTable 4 consistently report 118/733 vs 121/724 for the primary outcome, with OR 1.01, 95% CI 0.75 to 1.37, and P = .94; the point estimate lies within the CI, and the CI looks reasonable on the log-OR scale. jama_arleth_2024_oi_240147_1738… joi240147supp2_prod_1738701765.…

I did spot several issues, mostly in Supplement 2.

**1. Supplement 2, eTable 2, page 15: walk-in transport percentage appears incorrect.**
The liberal oxygen group reports “Walk-in” as 4/743 (5.3%). However, 4/743 = 0.54%, so this should be approximately **0.5%**, not 5.3%. joi240147supp2_prod_1738701765.…

**2. Supplement 2, eTable 2, page 15: vascular surgery percentage appears impossible.**
For surgery performed in the trauma resuscitation room, the liberal oxygen group reports “Vascular surgery” as 0/747 (1.1%). If the numerator is 0, the percentage should be **0.0%**. This is likely either a percentage typo or a numerator typo. joi240147supp2_prod_1738701765.…

**3. Supplement 2, eTable 7, page 20: traumatic brain injury subgroup percentage is inconsistent with the numerator/denominator and with the main manuscript.**
In eTable 7, the AIS < 3 subgroup for the liberal oxygen group is reported as 48/473 (9.2%). But 48/473 = 10.1%. The main manuscript’s Figure 4 reports the same row as 48/473 (10.1%), which appears correct. Therefore, the eTable 7 percentage is likely wrong. joi240147supp2_prod_1738701765.… jama_arleth_2024_oi_240147_1738…

**4. Supplement 2, eTable 1, page 14: several receiving trauma center percentages do not match the denominators.**
For the receiving trauma center rows, several percentages appear inconsistent. For example, 335/750 is 44.7%, not 44.4%; 348/758 is 45.9%, not 45.7%; 50/750 is 6.7%, not 7.2%; and 47/758 is 6.2%, not 6.7%. These look like table-arithmetic or manual-rounding errors. They do not appear to affect the main outcome results, but they are exactly the kind of errors your proposed AI-validation project could detect. joi240147supp2_prod_1738701765.…

**5. Supplement 2, eTable 10, page 24: denominator/percentage presentation is confusing and partly inconsistent.**
The row “Exclusion after randomization” is labeled N = 130, but the cells show 55 and 67, summing to 122. Also, the entries are written as 55/750 (45) and 67/758 (55). If the denominators are truly 750 and 758, the percentages would be 7.3% and 8.8%, not 45% and 55%. The 45% and 55% appear to describe the distribution among excluded patients, not within-group percentages. The secondary exclusion row has a footnote explaining that 2 patients had missing intervention data, but the first row’s N = 130 vs 122 discrepancy is not clearly explained in the table. joi240147supp2_prod_1738701765....

**6. Supplement 2, eTable 11, page 25: minor typographic error.**
The best/worst-case table reports “135//750 (18.0)” with a double slash. Numerically, 135/750 = 18.0%, so this appears to be only a typographical issue.
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**7. Protocol/SAP vs final manuscript: statistical software may need clarification.**
The SAP states that SAS version 9.4 would be used for all analyses.
joi240147supp1_prod_1738701765.... In the main manuscript text extraction, the final analysis appears to report use of R version 4.3.1. This is not necessarily a statistical-result error, but it is a protocol/SAP-to-reporting discrepancy that would be worth flagging in a validation workflow.

Overall, this paper is a good pilot example for your project. The major clinical/statistical conclusions seem internally consistent, but the supplements contain several table-level arithmetic and formatting issues. That supports your motivation: these are plausibly honest reporting or production errors, yet they are exactly the type of issue that reviewers and readers can miss, and AI-assisted checking could help surface them systematically.